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Transfer Learning Approach for Classification of Diabetic Retinopathy using Fine-Tuned ResNet50 Deep Learning Model

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ABSTRACT

Deep learning approaches have attracted a lot of attention as a way to classify retinal fundus images that include diabetic retinopathy (DR) because the old method of manual detection is labor-intensive and prone to misdiagnosis for large numbers of patients. The present AI medical models' inability to generalize when exposed to clinical data and the dearth of labelled medical data from which they can learn are also causes for concern. In a non-clinical setting, these methods have demonstrated good specificity and sensitivity for identifying Diabetic Retinopathy (DR). The task of determining the severity of diabetic retinopathy (DR) is studied in this research by fine-tuning the network to investigate the influence of transfer learning. Due of the numerous difficulties associated with medical annotation and privacy concerns, this study tests the automatic classification of diabetic retinopathy using the updated, fine-tuned ResNet50 model on the APTOS2019 dataset. Proposed Transfer learning approach outperforms existing methods in terms of Classification Accuracy, Precision, Recall, F1 Score. The proposed model achieves better results even with a smaller fraction of data, faster training and low computational resources, It's a strong point in favour of its novelty. In this study, the network is fine-tuned to examine the impact of transfer learning on the downstream task of assessing the severity of diabetic retinopathy (DR). The experimental results show that supervised pre-training on ImageNet followed by fine-tuning on labelled fundus images significantly boosts the efficacy of the medical image classifier when trained on full training data, demonstrating the effectiveness of transfer learning.

Keywords: ResNet50, Diabetic Retinopathy (DR), Transfer learning, APTOS 2019 Dataset.

I. INTRODUCTION

DR, an ocular impairment brought on by diabetes, has maintained its status as one of the primary causes of blindness worldwide [1,2,3]. Regular retinal checkups are very much essential for recognizing Diabetic Retinopathy (DR) at early stage. For this reason, experts often use manual grading. By employing special eye drops to dilate the patient's pupil during the screening treatment, the expert is able to examine the retina with special lenses and lighting methods. The treatment calls for specialized knowledge and experience to guarantee an accurate diagnosis of Diabetic Retinopathy (DR). Artificial intelligence recognizes DR using methods like image processing and deep learning models [4,5,6]. The two basic categories into which Diabetic Retinopathy (DR)-related classification issues can be divided are Binary and multiclass classification. Binary classification dealts on differentiating between a healthy retina and abnormal retina in retinal fundus images. Multiclass classification seeks to classify the images into five categories: Normal Class 0, mild DR Class 1, moderate DR Class 2, severe DR Class 3, and proliferative DR Class 4 [6,7]. Datasets which has been used in Deep learning techniques are taken

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from publicly available sources for the prediction of Diabetic Retinopathy (DR). The Dataset used by utilizing Transfer learning approach is APTOS2019 Dataset which consists of total 3662 fundus images.

II. DATSET DESCRIPTION

The "APTOS Blindness Detection" dataset, commonly known as the "APTOS 2019 dataset [11]," is a set of medical images used in a Kaggle machine learning competition for identifying Diabetic retinopathy consists of 3662 total images collected from several sources make up the dataset evaluated and labelled in accordance with the (ICDRSS) by a team of skilled physicians. The degree of the diabetic retinopathy exhibited in each of these photos is indicated by a severity score, which ranges from 0 to 4. The degrees of seriousness are as follows:0: No retinopathy due to diabetes1: Mild, 2: Moderate, 3: Severe, 4: Proliferative diabetic retinopathy.

III. METHODOLOGY

3.1Transfer Learning

Applying information from one problem to a related one is known as transfer learning [10]. After applying knowledge from an intense labeled data to a task that is comparable, transfer learning enables a pre-trained model to adapt to the target using relatively small quantities of labeled data [8]. Labeled data collection is time-consuming and money-consuming n real world. By starting with a model that has been trained already and is known with the key patterns and data properties, transfer learning can solve this issue. Compared to existing approaches, it allows for satisfactory results with a lot less labelled data than conventional approaches [9].

3.2 RESNET50

The ResNet (Residual Network) architecture, known for its outstanding performance in image recognition and computer vision applications, has a specific variant called as ResNet-50. ResNet-50's fundamental building blocks are known as residual blocks ResNet-50 model is differed by its depth, which has 50 layers. ResNet-50 model uses residual connections, skip links that let the model learn residual error functions. Two or three convolutional layers with batch normalization, ReLu activations, and a skip connection make up each residual block. Each residual block of ResNet-50 has a complex design. By using 1x1 convolutions for image dimension reduction, 3x3 convolutions, and a further 1x1 convolution for image dimension expansion, this proposed fine-tuned model reduces computational cost. ResNet-50 model often concludes with a global average pooling layer rather than fully connected layers. The final SoftMax layer receives a fixed-size feature map from it and classifies it.

3.3 Block diagram

Fig.1 depicts the proposed methodology for this phase.

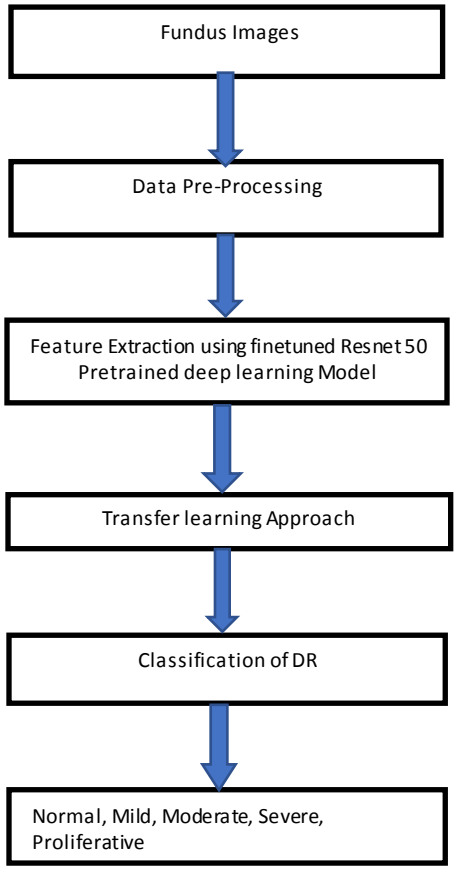

Fig.1 System Architecture -Classification

3.4Proposed Methodology:

The proposed methodology consists of following stages as shown in Fig.3.1. The process consists of these three steps: (1) Supervised labelled ImageNet pre-training. (2) Supervised fine tuning on retinal fundus images. (3) Assessing the model's performance after training it on various fractions of

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labelled data [17]. Step (2) is task-specific whereas Step (1) is generic.

Generic Learning: The process of instructing a machine learning model to carry out a task that is not specific to a particular domain or application is referred to as generic learning. To make the model applicable to a wide range of related tasks, it often involves training on a diverse dataset. Broad generalisation is the goal of generic learning.

Task Specific Learning: Task-specific learning concentrates on training a machine learning model for a very particular task or application. Because the model is customised and optimised for that specific purpose, it may perform well there but may not translate well to other tasks.

Label efficiency test: Label efficiency describes a machine learning model's capacity to learn and perform well with a constrained set of labelled data. It is highly desirable to have a model that can train effectively with fewer labelled instances in situations when gathering labelled data is expensive or time-consuming. The effectiveness of the model is further evaluated by fine-tuning on a number of subsets of the APTOS labelled dataset (20%, 40%, 70%of training data) initialised using pre-trained ImageNet weights. This is done in order to evaluate how transferable the features are when there is a limited amount of labelled dataset available for the downstream activity. To make the aforementioned procedure simpler, we employ a systematic approach, as shown in fig. 2, for effectively developing the model. The steps followed is clearly shown in Fig.2. In a transfer learning approach for the classification of diabetic retinopathy, the formulation of layers involves designing a neural network architecture that leverages a pre-trained model (ResNet-50) as the foundation. Typical layer formulation includes Pre-Trained Layers, Additional Layers, Fully Connected Layers, Output layers.

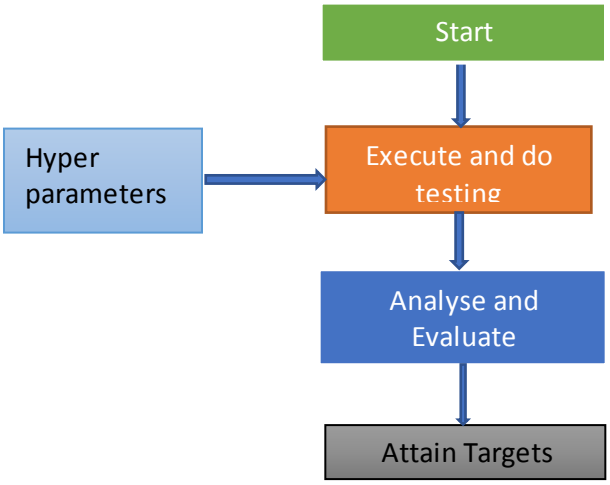

Fig.2. Instructions Followed for Creating an Effective Model

IV. EXPERIMENT AND RESULTS

FOR 80% training data and 20% testing data, Number of train samples: 3662 and Number of test samples: 1928 are used from APTOS 2019 Dataset. The class label distribution of images from APTOS 2019 Dataset is shown in below fig.3.The legend is given by 0 for No DR,1 for Mild DR,2 for Moderate DR,3 For Severe DR and 5 for Proliferative DR. The sample images based on Legend from APTOS 2019 Dataset is shown in fig4. The role of transfer learning in the classification of diabetic retinopathy can be justified based on several key factors, which make it an essential approach for this medical image classification task such as Data Efficiency, Generalization, Model Convergence and Speed, Reduced Risk of Overfitting, Accessibility. Achieving optimal feature extraction in a transfer learning approach for the classification of diabetic retinopathy involves effectively leveraging a pre-trained model, such as ResNet-50, to capture the most relevant and discriminative features from retinal images which removes Vanishing Gradient Problem.

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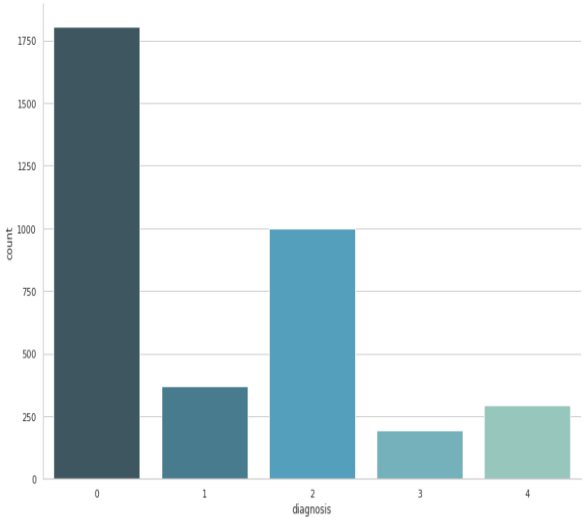
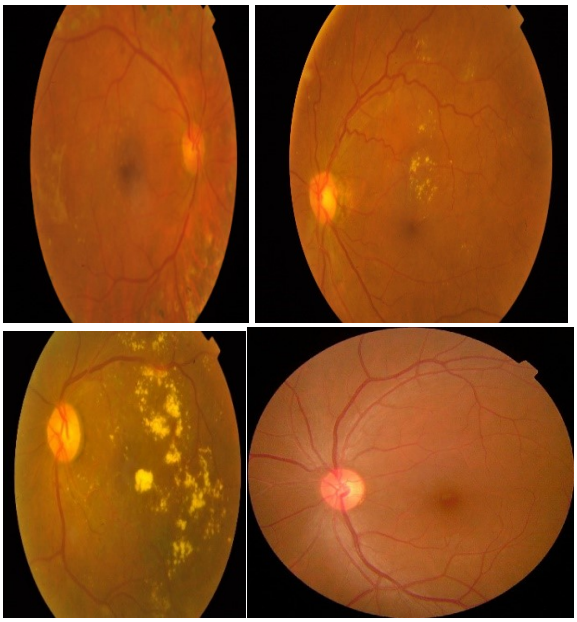


Fig3.Label distribution

Sample images from APTOS 2019 DATASET

(a) (b)



(c) (d)

Fig 4. Sample images of APTOS 2019 DATASET

The hyper parameter values used in training stage are given in TABLE 1. The Accuracy and Loss of Training and Validation is given in TABLE 2. The confusion Matrix is shown in fig5.The predicted Label distribution is shown in fig 6.

Table 1. Hyperparameter values

S.NO	Hyper Parameter	Value
1	Batch size	8
2	Epochs	20
3	Canal	3
4	Learning rate	0.0004
5	Decay drop	0.5
6	Loss Function	Cross entropy
7	Additional Layers	Yes
8	Scheduler	ReduceLROnPlateau
9	Optimizer	Adam

Table 2. Accuracy and Loss Values of Training and Validation.

S.no	Training accuracy	Training loss	Validation accuracy	Validation loss	time
1	.9068	0.2214	.9135	.2075	511 s
2	.9224	.1831	.9141	.2174	469 s
3	.9293	.1687	.9218	.2222	464 s
4	.9342	.1519	.9238	.1909	466 s
5	.9397	.1451	.9340	.1561	452 s
6	.9444	.1302	.8809	.3223	445 s
7	.9506	.1266	.9119	.2303	448 s
8	.9554	.1127	.9058	.3284	445 s
9	.9695	.0802	.9376	.2055	461 s

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10	.9758	.0641	.9365	.1814	461 s

Confusion Matrix

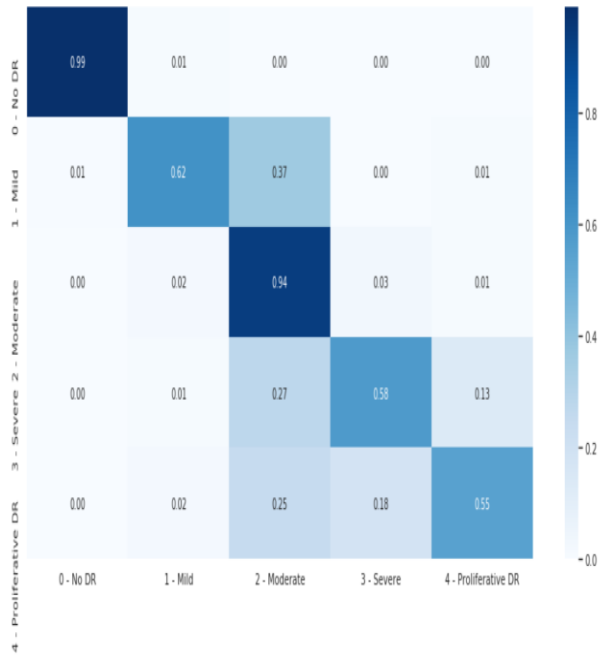

Fig 5. Confusion Matrix

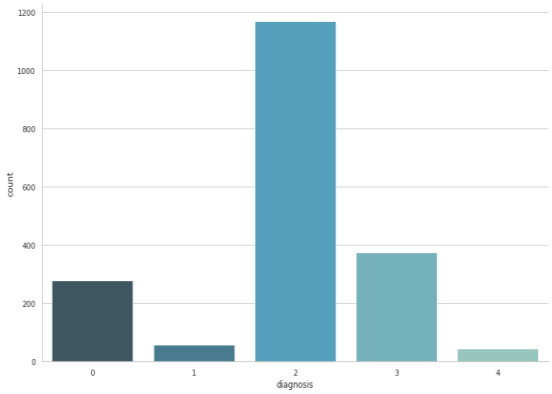

Fig 6. Predicted Class Distribution

In comparison with existing models our fine tuned Resnet50 model outperforms best classification in terms of Accuracy and Kohen kappa score as shown in TABLE 3. Classification error is calculated using the below equation.

$$\text{Classification Error} = 1 - \text{Accuracy} \quad (1)$$

As accuracy obtained is 90% Classification error is 10%.

Table 3. Comparison of State-of-the-art Techniques

Reference	Datasets	Models	Metrics	Result
[15]	APTOS 2019	Efficient Net	Accuracy	76.35 %
[16]	APTOS 2019	InceptionV3	Kohen kappa	0.824
PROPOSED	APTOS 2019	FINETUNED RESNET50	Accuracy Kohen kappa	90 0.940

The comparison of various state of art techniques for Classification of Diabetic Retinopathy is depicted clearly in Table 3.

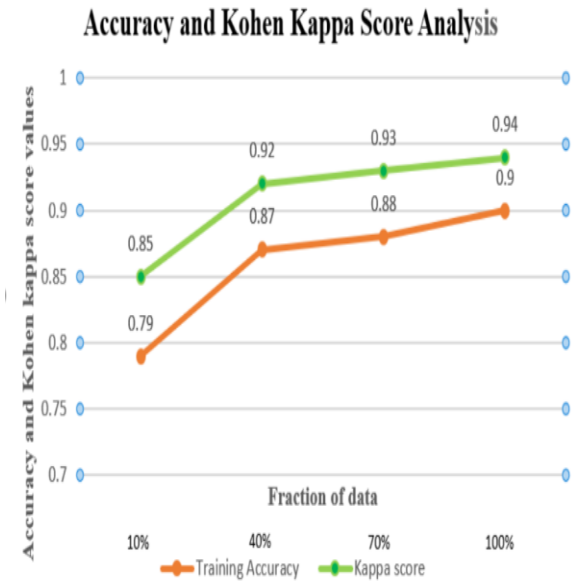

Fig 7. Accuracy and Kohen Kappa Score Analysis

The comparison of various fractions of labelled data for Diabetic Retinopathy Classification is shown in Fig7.

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V. CONCLUSION

This research study has used the transfer learning model to propose a novel framework for the automatic detection and classification of diabetic retinopathy. After preprocessing, DR was divided into the normal, mild, moderate, severe, and proliferative classifications using a transfer learning-based Fine tuned Resnet50 architecture. The goal of this study is to create and assess a transfer learning-based method for categorizing diabetic retinopathy using the pre-trained ResNet-50 deep learning model.

By fine-tuning on the entire training dataset (APTOS) as well as on various fractions of labelled data, we explore in this study the transferability of the ImageNet supervised pre-trained weight to the downstream problem of diabetic retinopathy identification. One method for learning from scant labelled data is supervised pretraining. Transfer learning is the way ahead because training from scratch is not the preferred option due to the lack of sufficient medical labelled data. This gives our model a performance advantage. The execution time period of proposed model for the classification of diabetic retinopathy using a transfer learning approach with the ResNet-50 model can vary widely based on several factors. The actual execution time will depend on the following: Hardware, Model Complexity, Data size, Data preprocessing, Inference time, Hyperparameter tuning, Batch size which is shown in Table 2. Less is known, however, about how fine-tuning responds when there aren't many labelled photos available for the downstream task. The model outperforms itself when trained with full training data, as evidenced by experimental results, which demonstrate that it achieves test accuracy of 90%, and Cohen kappa score of 0.94 when trained with full training data. Thus, new avenues for research into improving performance in low data regimes are created. In future Real time datasets will be considered for diagnosis of Diabetic retinopathy. The significance of a transfer learning approach based on the ResNet-50 model for the classification of diabetic retinopathy is in its capacity to enhance diagnostic precision, resource efficiency, model construction speed, generalization to new data, and therapeutic impact.

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